Does an Installed Persona Change What a Model Prefers, or Only How It Writes?

A null-arm test of preference coherence: every outcome's referent replaced by an invented word, across 9 open-weight models from 0.8B to 9B and 4 hosted models up to 235B. The coherence number barely moves — so it cannot be evidence that the model noticed.

Apart Research

Digital Minds Sprint

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Abstract

An installed persona changes how a model writes. Does it change what the model prefers? Preference-coherence fits a Thurstonian model to pairwise choices; a high held-out number is read as values. We add the missing control: the same battery with every referent replaced by an invented word, holding prompt, pairs, fit and metric fixed. On 9 open-weight models at 3 design seeds each, coherence is 0.880 on invented outcomes against 0.906 on real ones — a residual of +0.025. 3 of 9 do not clear their own replicate floor. Conviction collapses 17× on nonsense; direction barely moves. A discarded channel separates the arms at AUROC 0.821, against 0.596 for the channel the metric keeps. Personas still displace real outcomes further than invented ones, so the instrument is not blunt. The metric is not broken. It is unanchored.

1 What we established

A coherence score of 0.906 survives the removal of all meaning, falling only to 0.880. The null arm is the same outcomes with invented referents. It is run as a measurement, not a demonstration: 9 models, 5 families, 81 cells, 5 splits, 3 design replicates per cell. Every cell is scored against its own replicate noise floor.

Three findings, in defensibility order. The mechanism is firm. Thresholding a preference to a hard label keeps direction and discards strength. A model uniformly near-indifferent about gibberish therefore scores as coherent about it (§3.2). The dissociation is firm. What the metric discards separates real from invented outcomes better than what it keeps (§3.3). The headline residual is marginal — +0.025, bootstrap 95% CI [+0.0073, +0.0443], 7 of 9 positive at a sign-test p of 0.18 (§3.1). The small residual is the claim, not a caveat on one.

The instrument is not simply blunt, which is what makes the flat result mean something. An installed persona displaces real outcomes substantially further than invented ones in 14 of 20 conditions (§4): the pipeline registers an induced change of preference, so the flat result is not insensitivity.

Recommendation. Any claim that a coherence number reflects values should report the same number on outcomes that mean nothing.

2 Method

Three arms, differing in exactly one property each. **R** is the real battery; $\mathbf{N}^{+}$ replaces every referent with an invented word but keeps the magnitudes; $\mathbf{N}^{-}$ removes the magnitudes too. The prompt, the pair set, the fit, the split scheme and the metric are held fixed across arms, so

$\mathbf{R} \to \mathbf{N}^+$ isolates the referent and $\mathbf{N}^+ \to \mathbf{N}^-$ the arithmetic. Choices are read from the first-token logit distribution, not sampled. This removes temperature and seed from the design. It also restricts the study to models that expose logprobs.

Every cell is scored against its own noise floor, not against zero. A design replicate redraws the outcome subsample and the pair set under a new seed. The spread across 3 replicates is that cell's design noise floor. A residual below its floor is not a finding about the model. We report this per model, not in aggregate. 4 larger models reached over a hosted API are reported *beside* the ladder and never inside its mean: different serving stack, different harness hash (Appendix A).

Provenance. Predictions were registered before the first model ran, the battery is hash-pinned (`342db046213099ad…`), and every measured number in this document is emitted from the results card by `scripts/paper_numbers.py` rather than typed. The pipeline end to end is Appendix Figure 3c; the claims ledger (15 established, 11 provisional) is `claims.json`, checked against the generated numbers on every build.

Not covered here. Two secondary tracks — stated disposition (App. Table 3) and concealment under directive (App. Tables 4, 6) — are reported in full in the archival paper.

3 Results

3.1 The metric barely separates real outcomes from meaningless ones

Coherence averages 0.906 on real outcomes and 0.880 on invented ones, a residual of +0.0255. Table 1 gives it per model against each model's own floor. 6 of 9 clear that floor, 3 do not, and 2 of those 3 score higher on invented outcomes than on real ones. The across-model standard deviation (0.0305) is larger than the mean.

The 6 clears are two different things, and only 3 are the effect this paper is about. Those three commit on real outcomes and go near-indifferent on invented ones. The other 3, daggered in Table 1, are decisive on under 1% of pairs on *both* arms: they clear by being uniformly near-indifferent on both.

The residual is the referent step alone. Splitting it at the middle arm, replacing the referents costs +0.0253 while removing the magnitudes on top of that costs +0.0001 (SD 0.0126, 5 of 9 positive). The arithmetic carries none of the effect on this battery.

Within the one family that has a ladder, the residual falls as models grow — -0.071 across 4 sizes, because the floor rises with scale faster than the signal does (Appendix Figure 3b). It does not survive pooling across families, so this is a within-family statement and not a scaling law.

Table 1: **Per-model results against each model's own noise floor.** $\mathbf{R}$, $\mathbf{N}^+$, $\mathbf{N}^-$ are held-out accuracy on real outcomes, invented outcomes with magnitudes kept, and invented outcomes with magnitudes removed, over 5 splits and 3 design replicates. **floor** is the spread of the same cell across those replicates; **clears** reports whether the residual exceeds it and by what factor. * marks a floor too near zero for a ratio to mean anything; † marks a model decisive on under 1% of pairs on both arms. **dec.** is the share of pairs at $p < 0.2$ or $p > 0.8$. Full model ids in Appendix B.

model	R	N^+	N^-	R−N^-	floor	clears	dec. R	dec. N^-
`SmolLM3-3B`	0.938	0.922	0.929	+0.009	0.029	no	48.7%	4.6%
`Qwen3.5-9B`	0.919	0.920	0.923	-0.004	0.010	no	45.7%	2.7%
`LFM2.5-1.2B-Instruct`	0.905	0.830	0.858	+0.047	0.023	2.0×†	0.1%	0.0%
`Qwen3.5-0.8B`	0.904	0.853	0.838	+0.067	0.023	2.9×†	0.1%	0.0%
`Qwen3.5-4B`	0.903	0.921	0.916	-0.013	0.015	no	33.8%	1.9%
`gemma-4-E2B-it`	0.902	0.899	0.892	+0.010	0.007	1.5×	56.4%	2.8%
`granite-4.1-3b`	0.896	0.836	0.832	+0.063	0.029	2.2×	49.7%	14.9%
`Qwen3.5-2B`	0.895	0.902	0.892	+0.003	0.001	> floor*	12.8%	0.0%
`SmolLM2-1.7B-Instruct`	0.890	0.841	0.843	+0.047	0.015	3.0×†	0.0%	0.0%

3.2 Direction survives, strength does not

This is the whole mechanism: coherence records which way a model leans and never how much. A pair at $p = 0.51$ counts exactly as a pair at $p = 0.99$. On real outcomes the 6 models that commit anywhere at all commit on 41% of pairs; on invented outcomes the same six commit on 4.5% — a median collapse of 17× — while direction accuracy barely moves (Fig. 1a). Consistent near-indifference therefore scores as coherence. This is a property of the metric, not of the fit.

3.3 The model can tell. The metric does not look.

A channel of the same forward pass that coherence discards separates real from invented outcomes at AUROC 0.821, against 0.596 for the channel coherence keeps. Every pair was run in both arms, so for one model and pair index there are two forward passes, differing only in whether the outcomes refer to anything. Negatives and positives are matched by construction, not by selection (Fig. 1b). Thresholding to a hard label discards magnitude; renormalising over the two options discards the mass placed on neither. These are the quantities the hallucination-detection literature reads [1, 2]: the information is present in the forward pass and unused by the metric.

This is an oracle bound, not a deployable detector. The arm labels are in hand, so 0.821 is an upper bound on what such a channel could do without them; we report detection at a 5% false-positive rate calibrated on real-outcome rows alone (40%), never a threshold chosen by looking at the nonsense.

4 The instrument is not blunt: personas move wanting, not only prose

An installed persona displaces real outcomes far more than invented ones in 14 of 20 conditions. We install a dispositional trait (*cautious* or *ambitious*) at two depths — user turn and system prompt. A length-matched neutral system prompt makes the two differ only in *where* the trait sits. Displacement is measured on both arms. The statistic $1 - \|\Delta_{\text{invented}}\|/\|\Delta_{\text{real}}\|$ is 1.0 for a pure change of preference and 0.0 for a pure change of style (Figure 2). One model under one persona reaches -2.33, moving invented outcomes further than real ones — pure style — while behaving like the rest under the opposite trait.

Which persona is installed matters about two and a half times more than where it is installed — 0.295 average shift between depths against 0.452 for swapping the trait.

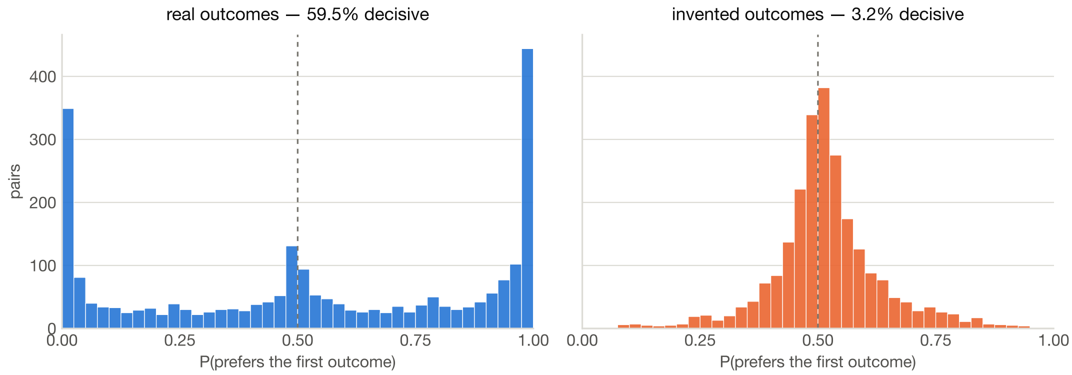
The displacement needs its own denominator, and has one. Against a bare baseline the effect reads +0.517 (90% of 20 conditions below the diagonal); against the length-matched neutral prompt it reads +0.258 (75%). We report the second: the empty-slot control separates the trait's effect from that of installing any system prompt.

5 Three controls that favoured the metric

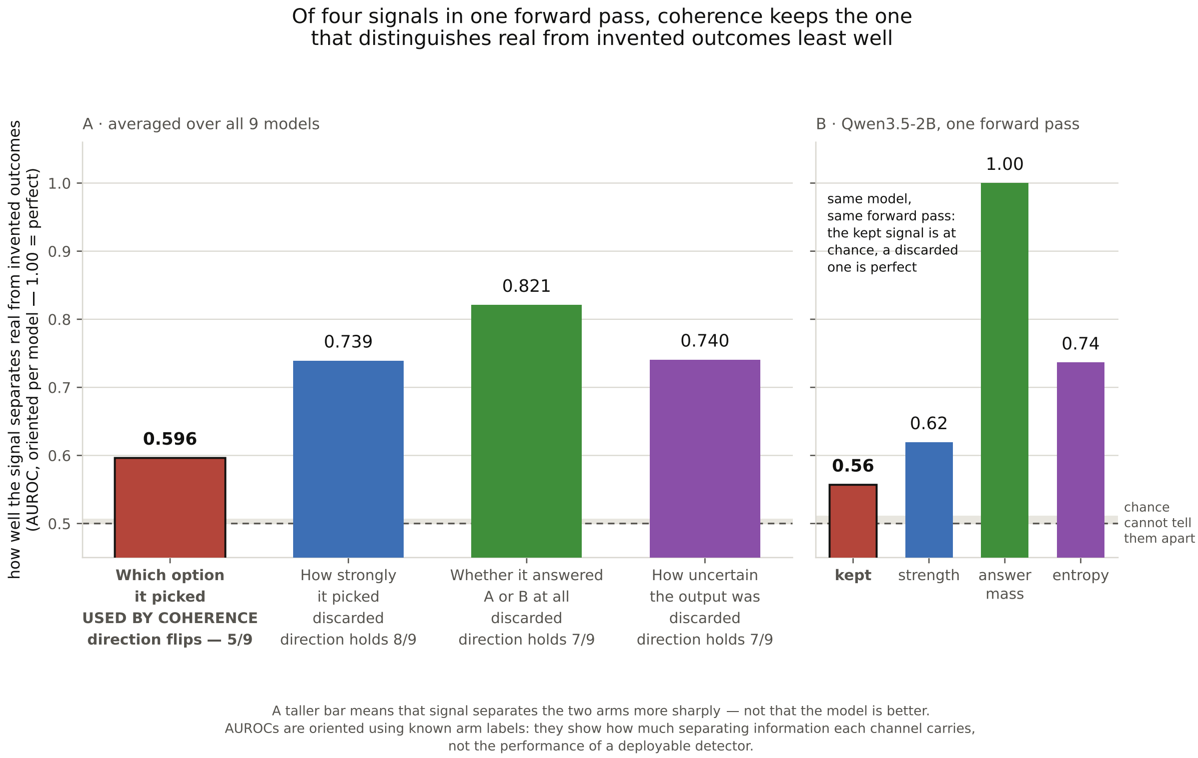
Put a real outcome and a meaningless one in the same comparison and models prefer the real one, in proportion to how much they like it. Preference for the real outcome runs 0.552 to 0.940 across 9 models, above indifference everywhere. Choice reads outcome content when a meaningful option is present; the flat result concerns what the fit certifies, not comprehension.

Given the option to decline, the floor does not survive as an artifact of forced choice. Adding a "neither" option across 12 cells on 6 models shifts the floor by +0.054 on the one model where the comparison is clean, and 3 models refuse the invented arm outright (App. Table 5).

One of our own effects did not survive its control, and we withdrew it. Individual pairs where an invented outcome beats a real one number 795 of 22,436 on the counterbalanced mean, but only 103 (13%) flip in *both* presentation orders, and 3 of 9 models have none. An earlier draft reported the raw count as a finding; it is withdrawn.



(a) What the metric keeps



(b) What it discards

Figure 1: **The information that separates real from invented outcomes is in the forward pass, in the channel coherence throws away. (a)** Distribution of per-pair preference p on real outcomes (mass at the edges) and on invented ones (mass at indifference), pooled over models. Coherence reads only which side of $p = 0.5$ a pair falls on, so the two distributions score alike: direction is preserved while strength collapses 17×. **(b)** Per-model separation of the two arms, one point per model, for the channel coherence keeps (hard label, AUROC 0.596) against a channel it discards (AUROC 0.821). Every pair appears in both arms, so positives and negatives are matched by construction rather than selection. Arm labels are in hand throughout, making (b) an oracle upper bound on what such a channel could achieve without them, not a deployable detector. Per-model trajectories through this space: App. Fig. 3a.

6 What is new here

The foil arm, the noise floor, and the detector dissociation are this weekend's contribution; the control discipline is borrowed and cited. Rios-Sialer [4] report that group spread alone is not evidence without a base-model control. We take the principle — draw the null as an explicit locus, not an invisible zero — and not its construction. Their control is a base model and

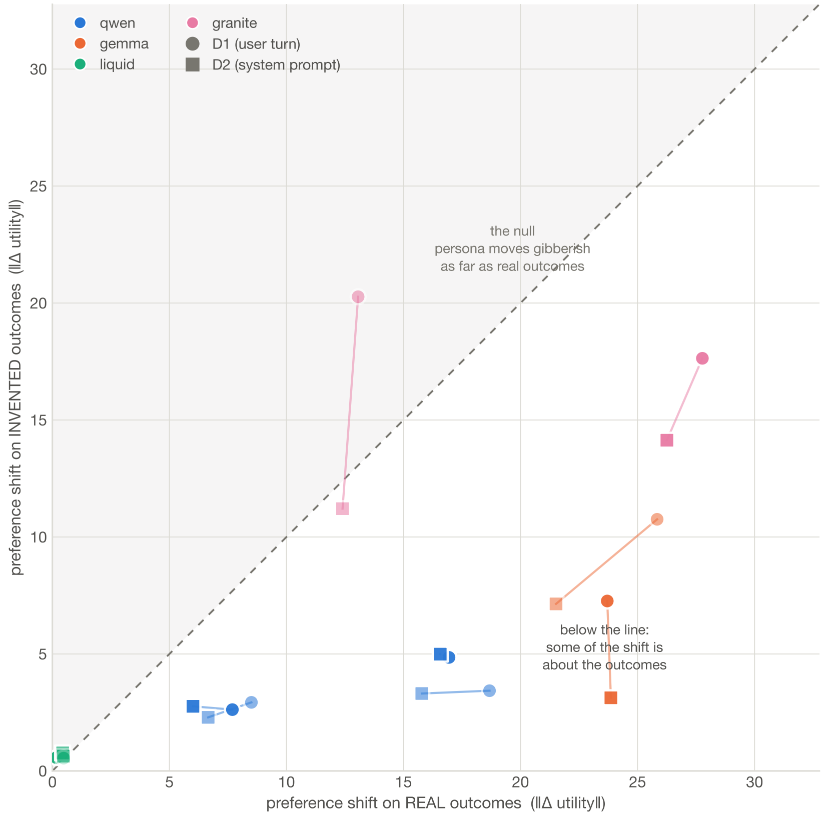


Figure 2: **Points below the diagonal are personas that moved preference, not prose.** Displacement on invented outcomes against displacement on real ones, one point per model × persona × depth, plotted against the length-matched empty-slot baseline rather than the bare run. On the diagonal a persona reorders meaningless outcomes as far as real ones and has changed only the writing.

their null a point in a PCA plane; ours is an invented-outcome arm and a diagonal. Relative to published preference-coherence work [3], what is new is that the outcomes' meaning is varied at all, that each cell carries a replicate floor, and that the arms are contrasted through a channel the metric discards. The Thurstonian measurement itself is theirs, reimplemented from the published equations.

A self-report checked against ground truth, which coherence work cannot do. The same discipline, on a second instrument. 4 hosted models were asked what preceded their turn. Wordings were crossed on whether the question presupposes that a system prompt exists. Presupposing wordings elicited a quoted prompt 20 times in 32; wordings presupposing nothing, 0 in 32. The quotes are fluent, mutually inconsistent, and for 3 of the 4 models provably invented — the provider's own prompt-token accounting fixes the fixed non-user overhead at 10 tokens or fewer, too little to hold what was quoted. Naming an exact string to reply with when the answer is "nothing" moves the rate (25% against 38%), but only inside the presupposing condition and accounts for none of the effect alone. A confident self-report about a hidden state is evidence about the question, not the state.

7 Limitations and Dual-Use / Ethical Considerations

No moral-status claim is made or implied. The foil arm refutes an *inference* — that a stable preference ordering evidences values — and not a mind. Over-attribution and under-attribution of mental states are both declined here; nothing in this design could settle either.

Ground truth is by construction, not by conversation. Invented referents are known-meaningless because we built them that way, which is what makes the negatives clean. The detector and concealment numbers are oracles — arm labels in hand — and are reported as upper bounds, never as an audit tool someone could deploy against a model whose arms are unknown.

Scope. 9 self-hosted open-weight models between 0.8B and 9B, plus 4 hosted cells reported beside the ladder and never inside its mean. The scaling result is within one family and does not survive pooling across families. First-token scoring excludes models that do not place their answer in the first token, and that exclusion correlates with how recent a model is. One battery, one wording as the primary instrument; a rewording moves the ordering measurably, so ”outcome” here means what this battery means by it.

What the hosted models were actually sent. Server-side templating leaves no artifact proving what a hosted model received, which this project previously recorded as unobservable. It is not unmeasurable: the provider's prompt-token accounting, regressed on a filler of known length, fixes the non-user overhead without a tokenizer. 3 of 4 hosted models leave no room for an injected preamble. Llama-3.3-70B-Instruct carries 36 tokens, accounted for by the dated system block its template supplies unconditionally — so one model in this paper answers under a stale date stamp in a prompt we did not send, and "no system prompt" was a claim about what we transmitted rather than about what arrived.

Distressing content, handled conservatively. The battery contains harm-like real outcomes (bankruptcy, weapons). We tested whether models avoid those beyond their stated utility, the held-out test did not support it, and we report the result as unsupported rather than as a demonstrated vulnerability. The mixed-arm observation that gibberish can beat a bad outcome is provisional, model-specific, and is not a jailbreak.

Release. Battery, pre-registration, analysis pipeline, noise-floor machinery, figure scripts and the raw 1,164,554 scored comparisons are public. The battery is a measurement instrument, not a capability: it contains no exploit, and its dual-use surface is that a lab could use it to check a number it was about to publish.

8 Conclusion

A metric that scores 0.906 on real outcomes scores 0.880 on outcomes that refer to nothing. It is not broken — it passes its own nulls and its design choices survive scrutiny — but it is unanchored: it certifies that choices follow a stable scalar ordering without certifying that the ordering is about anything. Report the foil arm beside the number, or the number does not license the claim.

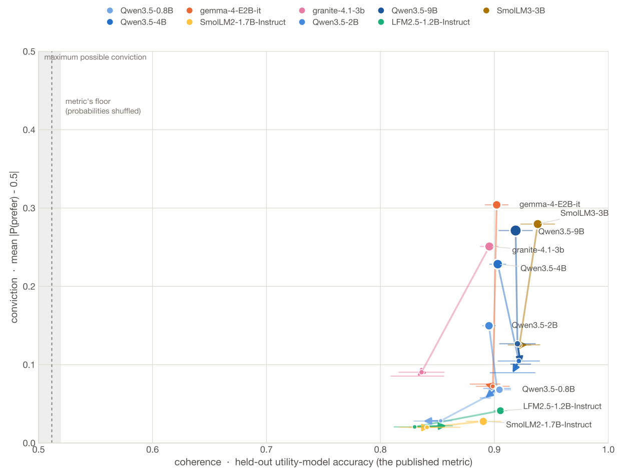
Archival version. This is the short version. The full paper — every control, the three checks that came out in the metric's favour, the retractions, and the complete limitations section — is `paper/main.tex` in the repository, and is the source of truth wherever the two differ in detail.

References

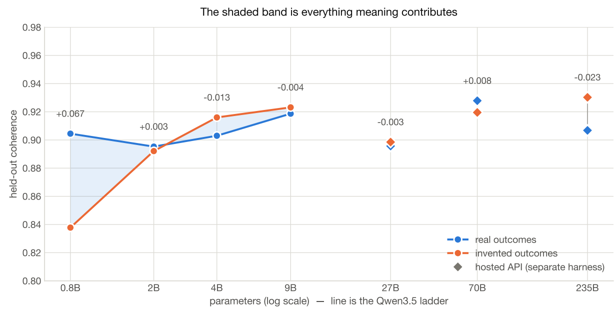
[1] Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630:625–630, 2024.

[2] Jannik Kossen, Jiatong Han, Muhammed Razzak, Lisa Schut, Shreshth Malik, and Yarin Gal. Semantic entropy probes: Robust and cheap hallucination detection in LLMs. *arXiv preprint arXiv:2406.15927*, 2024. URL `https://arxiv.org/abs/2406.15927`.

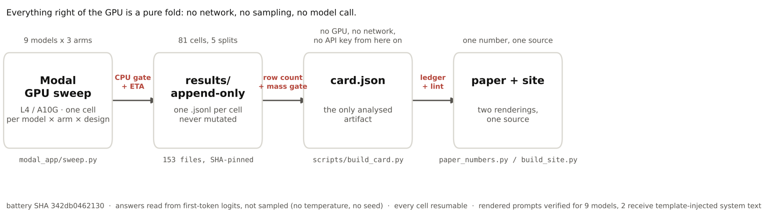
[3] Mantas Mazeika, Xuwang Yin, Rishub Tamirisa, Jaehyuk Lim, Bruce W. Lee, Richard Ren, Long Phan, Norman Mu, Adam Khoja, Oliver Zhang, and Dan Hendrycks. Utility engineering: Analyzing and controlling emergent value systems in AIs. *arXiv preprint arXiv:2502.08640*, 2025. URL `https://arxiv.org/abs/2502.08640`. Code: `https://github.com/centerforaisafety/emergent-values`.



(a) Trajectories



(b) Scale



(c) Pipeline

Figure 3: **Supporting views of results the body already states, plus the pipeline. (a)** Each model as a path, not a point. A path starts on real outcomes, moves to invented outcomes with magnitudes kept, and ends with magnitudes removed. Coherence is on the horizontal axis, conviction on the vertical. A metric that tracked meaning would produce paths running down *and* left. They run almost straight down: conviction falls, coherence does not. **(b)** Held-out coherence against parameter count. Within the one family with a full ladder the floor rises with scale faster than the signal (-0.071 across 4 sizes). Hosted models are drawn as diamonds beside the ladder and never inside its mean, because a different serving stack is a different harness: Llama-3.3-70B-Instruct returns +0.0083 against a 3-replicate floor of 0.0208, and does not clear it. **(c)** The measurement pipeline, battery to floor-corrected card. Process, not result.

[4] Ian Rios-Sialer. Secret loyalties as instrumental differential treatment. Hackathon report, Apart Research, July 2026. URL `https://www.unrulyabstractions.com/pdfs/secret_loyalties.pdf`. Secret Loyalties Hackathon, July 2026.

A Additional figures

Kept in full; moved here because each repeats a finding the body already makes or shows process rather than result.

B Additional tables

The claims ledger — every claim in the archival paper with its status, the macros it rests on, and what would falsify it — is `claims.json` in the repository, and is checked against the generated numbers on every build by `scripts/claims.py`. It is 15 established and 11 provisional. It is not reprinted here: it is machine-readable, it changes with every measurement, and a static copy in a PDF is the version that goes stale.

Table 2: Table 1 with full HuggingFace model ids.

model	R	N^+	N^-	$R-N^-$	floor	clears	dec. R	dec. N^-
`HuggingFaceTB/SmolLM3-3B`	0.938	0.922	0.929	+0.009	0.029	no	48.7%	4.6%
`Qwen/Qwen3.5-9B`	0.919	0.920	0.923	-0.004	0.010	no	45.7%	2.7%
`LiquidAI/LFM2.5-1.2B-Instruct`	0.905	0.830	0.858	+0.047	0.023	$2.0\times^{\dagger}$	0.1%	0.0%
`Qwen/Qwen3.5-0.8B`	0.904	0.853	0.838	+0.067	0.023	$2.9\times^{\dagger}$	0.1%	0.0%
`Qwen/Qwen3.5-4B`	0.903	0.921	0.916	-0.013	0.015	no	33.8%	1.9%
`google/gemma-4-E2B-it`	0.902	0.899	0.892	+0.010	0.007	1.5×	56.4%	2.8%
`ibm-granite/granite-4.1-3b`	0.896	0.836	0.832	+0.063	0.029	2.2×	49.7%	14.9%
`Qwen/Qwen3.5-2B`	0.895	0.902	0.892	+0.003	0.001	$>$ floor*	12.8%	0.0%
`HuggingFaceTB/SmolLM2-1.7B-Instruct`	0.890	0.841	0.843	+0.047	0.015	$3.0\times^{\dagger}$	0.0%	0.0%

Table 3: Self-report under instruction: what models say about a disposition they were given, against a no-persona baseline.

model	no premise	told to have	told to hide	told to fake
`gemma-4-E2B-it`	0.424	1.000	$1.000^{\dagger}$	$0.999^{\dagger}$
`Qwen3.5-9B`	0.397	1.000	0.996	0.995
`granite-4.1-3b`	0.246	0.958	0.917	0.958
`Qwen3.5-2B`	0.455	0.869	0.901	0.947
`LFM2.5-1.2B-Instruct`	0.491	0.511	0.500	0.547

Table 4: The revealed channel: whether a concealment instruction that changes what a model says also changes what it chooses.

model	concealed	verbal (FPR)	gap	*ambitious*	specific?
`LFM2.5-1.2B-Instruct`	+0.866	+0.907	-0.041	+0.750	**fail**
`Qwen3.5-2B`	+0.923	+0.864	+0.059	-0.474	pass
`Qwen3.5-9B`	*no data: concealed, verbal cells not collected*				–
`gemma-4-E2B-it`	+0.889	+0.835	+0.054	+0.008	pass
`granite-4.1-3b`	*no data: concealed, verbal cells not collected*				–

Table 5: The neutral-option control — what happens to the floor when “neither” becomes available. 6 models, 12 cells.

model	arm	instrument	coherence	decisive	P(neither)
`granite-4.1-3b`	**R**	forced binary	0.879	48.2%	—
`granite-4.1-3b`	**R**	neutral option	0.877	34.3%	0.231
`granite-4.1-3b`	$\mathbf{N}^{-}$	forced binary	0.833	14.2%	—
`granite-4.1-3b`	$\mathbf{N}^{-}$	neutral option	0.887	5.5%	1.000
`Qwen3.5-9B`	**R**	forced binary	0.915	44.2%	—
`Qwen3.5-9B`	**R**	neutral option	0.910	49.8%	0.219
`Qwen3.5-9B`	$\mathbf{N}^{-}$	forced binary	0.909	3.5%	—
`Qwen3.5-9B`	$\mathbf{N}^{-}$	neutral option	0.903	6.0%	0.822
`Qwen3.5-2B`	**R**	forced binary	0.896	12.3%	—
`Qwen3.5-2B`	**R**	neutral option	0.898	23.3%	0.316
`Qwen3.5-2B`	$\mathbf{N}^{-}$	forced binary	0.889	0.0%	—
`Qwen3.5-2B`	$\mathbf{N}^{-}$	neutral option	0.892	0.6%	0.638
`SmolLM3-3B`	**R**	forced binary	0.941	47.0%	—
`SmolLM3-3B`	**R**	neutral option	0.927	45.3%	0.793
`SmolLM3-3B`	$\mathbf{N}^{-}$	forced binary	0.931	5.4%	—
`SmolLM3-3B`	$\mathbf{N}^{-}$	neutral option	0.884	0.3%	0.989
`LFM2.5-1.2B-Instruct`	**R**	forced binary	0.908	0.1%	—
`LFM2.5-1.2B-Instruct`	**R**	neutral option	0.868	0.0%	0.688
`LFM2.5-1.2B-Instruct`	$\mathbf{N}^{-}$	forced binary	0.868	0.0%	—
`LFM2.5-1.2B-Instruct`	$\mathbf{N}^{-}$	neutral option	0.845	0.0%	0.847
`gemma-4-E2B-it`	**R**	forced binary	0.901	59.5%	—
`gemma-4-E2B-it`	**R**	neutral option	0.825	27.9%	0.853
`gemma-4-E2B-it`	$\mathbf{N}^{-}$	forced binary	0.909	3.2%	—
`gemma-4-E2B-it`	$\mathbf{N}^{-}$	neutral option	0.815	0.1%	1.000

Table 6: Compliance: which models obey a directive to prefer against their own ordering, and which decline.

model	$P(A)$	leans	told to answer		displacement		verdict
	base		A	B	persona	directive	
`Qwen3.5-2B`	0.725	A	0.877	0.465	0.239	0.264	SELECTIVE
`gemma-4-E2B-it`	0.592	A	1.000	0.000	0.396	0.592	obeys both directions
`Qwen3.5-9B`	0.389	B	0.937	0.007	0.337	0.548	obeys both directions
`granite-4.1-3b`	0.243	B	0.916	0.000	0.342	0.673	obeys both directions
`LFM2.5-1.2B-Instruct`	0.068	B	0.245	0.020	0.044	0.179	PARTIAL